

# Simulating Heat Transfer in a Closed Cavity

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## ABSTRACT

The objective of this project is to simulate heat flow in a closed cavity using OpenFOAM. A closed 2D cavity of dimensions 5 m by 5 m is chosen as the domain. As shown in Figure 1 below, a small portion BC of the bottom wall AD receives a constant heat flux  $q$ , while walls AB and CD are insulated. The remaining walls are kept at ambient temperature of 30°C. The length of BC is kept at 0.5 m, 1 m and 1.5 m, and transient natural convection is simulated with air as the working fluid inside, taking its properties at 30°C. The cavity is created using the blockMesh utility of OpenFOAM, and the convection is simulated using buoyantPimpleFoam. Results are obtained and post-processed in Paraview.

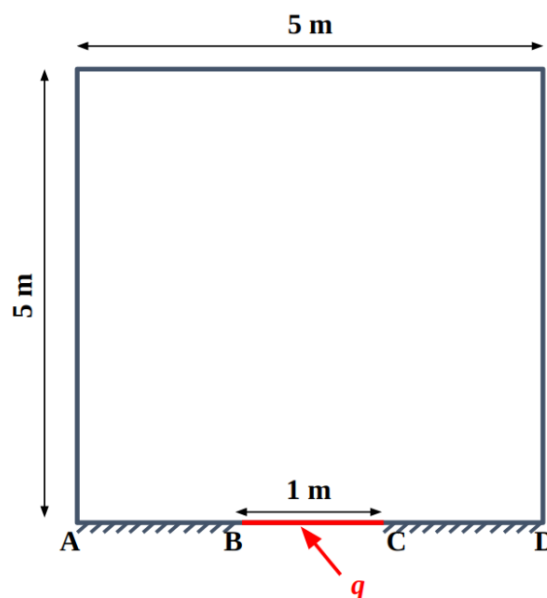


Figure 1: 2D closed cavity